Noisy Rational Expectation Equilibrium

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Abstract

I worked on the standard model first, and then extend it to explain possible asset pricing puzzles. The goal is to explain Lu's puzzle and term structure of dividend stripes.

Homework
Solution

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1. Standard Model

Let us start with the most basic model, and then see the implication of this model when generalized, e.g. heterogenous agents with different risk aversion level, market power(market microstructure style).

1.a. Assets

- Two period $t = 0, 1$
- One riskless asset with deterministic return r >1 with perfect elastic supply.
- N risky asset with payoffs $f \sim N(\mu, \Sigma)$.
- Risky assets are scarse and supply $\bar{x} + x$, where $x \sim N(0, \sigma_x^2 I)$

Can we make the μ bayesian style? Say, μ is also a random variable. We have good results when payoff f is in discrete probability space. In multivariate normal settings, the law of one price and pricing kernal is not very obvious? What happen if probability state is discrete?== Agent

Agent i maximizes expected Constant Absolute Risk Averse(CARA) utiltiy function over second period payoff,

$$E_i[U(X)] = E_i[-\exp(-\rho X)]. \tag{1}$$

In this model, all payoffs are linear combinations of the payoffs of one riskless asset and N risky asset. So X follows multivariate normal distribution. Maximizing expected utility becomes maximizing mean variance trade-off $E_i(X) - \frac{\rho}{2} VAR_i(X)$. Agent i buys q_i risky assets and spends $q_i{}'p$ on risky assets with payoff $q_i{}'f$. The remaining wealth $W_{0,i} - q_i{}'p$ goes to riskless asset with payoff $\big(W_{0,i} - q_i{}'p\big)r$. The portfolio choice problem by maximizing expected utility $E_i[U_i]$ is

$$\max_{q_i} \ \big(W_{0,i} - q_i{}'p\big)r + q_i{}'E_i[f] - \frac{\rho}{2} q_i{}'VAR_i(f)q_i \tag{2}$$

and the optimal portfolio choice is

$$q_i^* = \frac{1}{\rho}\left[VAR_i(f)\right]^{-1}(E_i[f] - pr). \tag{3}$$

- $\rho = -\frac{U''}{U'}$: absolute risk aversion parameter. Higher risk averse level detains willingness to hold risky asset.
- $\left[VAR_i(f)\right]^{-1}$: the precision of the inference of f. Uncertainty detains willingness to hold risky asset.
- Agent takes price p as given, which determines all three components $\left[VAR_i(f)\right]^{-1}$, $E_i[f]$ and pr.

1.b. Information Set

- Agent i forms beliefs and expectations based on information $\mathcal{I}_i$ she observed.
- $\mathcal{I}_i = \{s_i, p\}$ both public and private information
- s_i: private signal observed by agent i
- p: price vector observed by every agent
- $E[f \mid \mathcal{I}_i]$, $Var[f \mid \mathcal{I}_i]$: Agent i uses information $\mathcal{I}_i$ to form expectations.
- f is realized but unobservable. Agent needs to infer f.

Agent i has received noisy private signal

$$s_i = f + \varepsilon_i, \ \varepsilon_i \underset{IID}{\sim} N\big(0, \Sigma_{\varepsilon,i}\big), \tag{4}$$

the expectation formed based on private information is

$$E[f \mid s_i] = s_i, Var[f \mid s_i]^{-1} = \Sigma_{\varepsilon,i}^{-1}. \tag{5}$$

Agent i can also extract information from price signal p, by assuming the price is reflecting f and x in linear form,

$$p = A + Bf + Cx, \tag{6}$$

\sum_i W_{0,i} doen't show up in pricing equation because of CARA assumption. The expectation formed based on public information is

$$E[f \mid p] = B^{-1}(p - A), Var[f \mid p]^{-1} = \left(B^{-1}C(B^{-1}C)^{-1}\right)^{-1} := \Sigma_p^{-1}. \tag{7}$$

Prior belief says

$$f \sim N(\mu, \Sigma) \tag{8}$$

the pior expectations are

$$E[f] = \mu, Var[f]^{-1} = \Sigma^{-1}. \tag{9}$$

Doing Bayesian updates, the total precision is

$$Var[f \mid \mathcal{I}_i]^{-1} = Var[f]^{-1} + Var[f \mid s_i]^{-1} + Var[f \mid p]^{-1} = \Sigma^{-1} + \Sigma_{\varepsilon,i}^{-1} + \Sigma_p^{-1} \tag{10}$$

and the expected payoff is

$$E[f \mid \mathcal{I}_i] = Var[f \mid \mathcal{I}_i]\big(\Sigma^{-1}\mu + \Sigma_{\varepsilon,i}^{-1}s_i + \Sigma_p^{-1}B^{-1}(p - A)\big) \tag{11}$$

1.c. Equilibrium Condition

For simplicity, let's work on the simplest case where population is uniformly distributed.

- A continuum of investors indexed by $i \in [0, 1]$. Each takes prices as given, and has an initial wealth $W_{0,i}$.

The investor's distribution determines how to aggregate from micro to macro.Assume agent distribution is uniform.

1.d. Stochastic discount factor

1.e. Market Power

If the number of investor is finite, $Pr(i) > 0$.

If the number of investor is infinite, then $Pr(i) = 0$

If we can use probability tools to unify markets with market power together, using Convergence Analysis.

By creating A "probability"